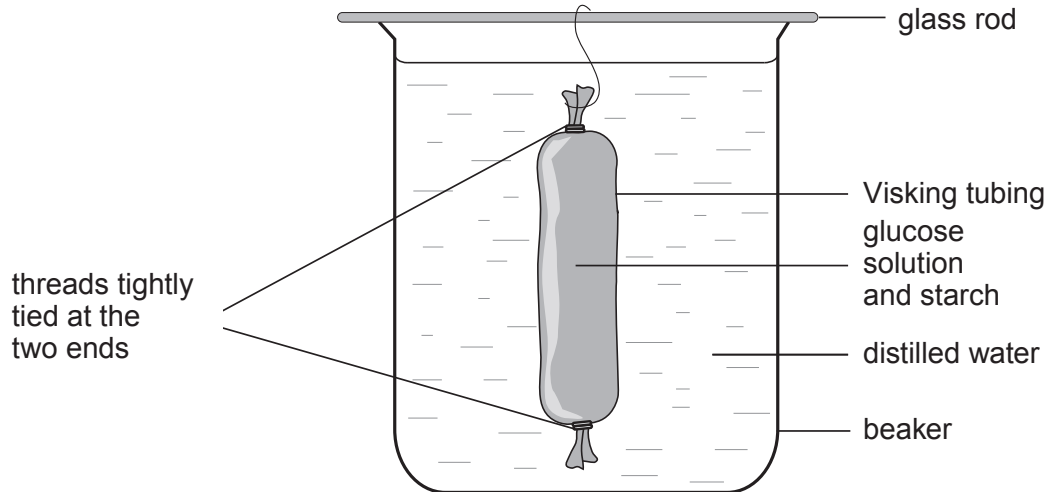


Answer **all** questions.

1. A student set up the following experiment to show how diffusion of digested foods occurs in the small intestine. The Visking tubing contained starch and a solution of glucose with a concentration of 100 g/dm^3 . This was placed in a beaker containing distilled water at 20°C and left for 2 hours.



- (a) (i) Explain what is expected to happen to the starch and the glucose in the Visking tubing during the experiment. [4]

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- (ii) Describe **two** chemical tests the student should carry out on the contents of the beaker to confirm your answer to (a)(i) and state the expected results. [4]

1.

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2.

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(b) Explain how each of the following changes would affect the diffusion process in this experiment.

(i) Carbohydrase is added to the contents of the Visking tubing. [2]

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(ii) The investigation is carried out at 35 °C. [2]

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(iii) The experiment is carried out with a solution of glucose of 100 g/dm³ concentration replacing the distilled water in the beaker. [2]

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(c) State **one** of the limitations of using Visking tubing as a model intestine. [1]

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(d) Explain **two** adaptations of the small intestine for the absorption of digested food. [4]

1.

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2.

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